

EXPERIMENT - 8

TO PREPARE AND COLLECT AMMONIA GAS AND STUDY ITS PROPERTIES.

APPARATUS REQUIRED

- | | |
|------------------------------|-------------------------|
| a) <u>Round bottom flask</u> | b) <u>Delivery tube</u> |
| c) <u>Cork</u> | d) <u>Gas jar</u> |
| e) <u>Burner</u> | f) <u>stand</u> |
| g) <u>Wire gauze</u> | h) <u>Water trough</u> |

CHEMICAL REQUIRED

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|----------------------------------|------------------------------|
| i. <u>Ammonium chloride</u> | ii. <u>Calcium hydroxide</u> |
| iii. <u>Copper Sulphate soln</u> | iv. <u>Ferric chloride</u> |
| v. <u>Mercurous nitrate</u> | vi. <u>Nessler's reagent</u> |
| vii. <u>Litmus paper</u> | viii. <u>Phenolphthalein</u> |
| ix. <u>Conc. HCl</u> | x. _____ |

THEORY

Ammonia gas can be obtained by heating ammonium salt with alkali. In laboratory ammonia gas is prepared by heating paste of ammonium chloride (sal ammoniac) and calcium hydroxide (slaked lime).



Slaked lime should be used in excess because excess OH^- ion given by slaked lime favors the

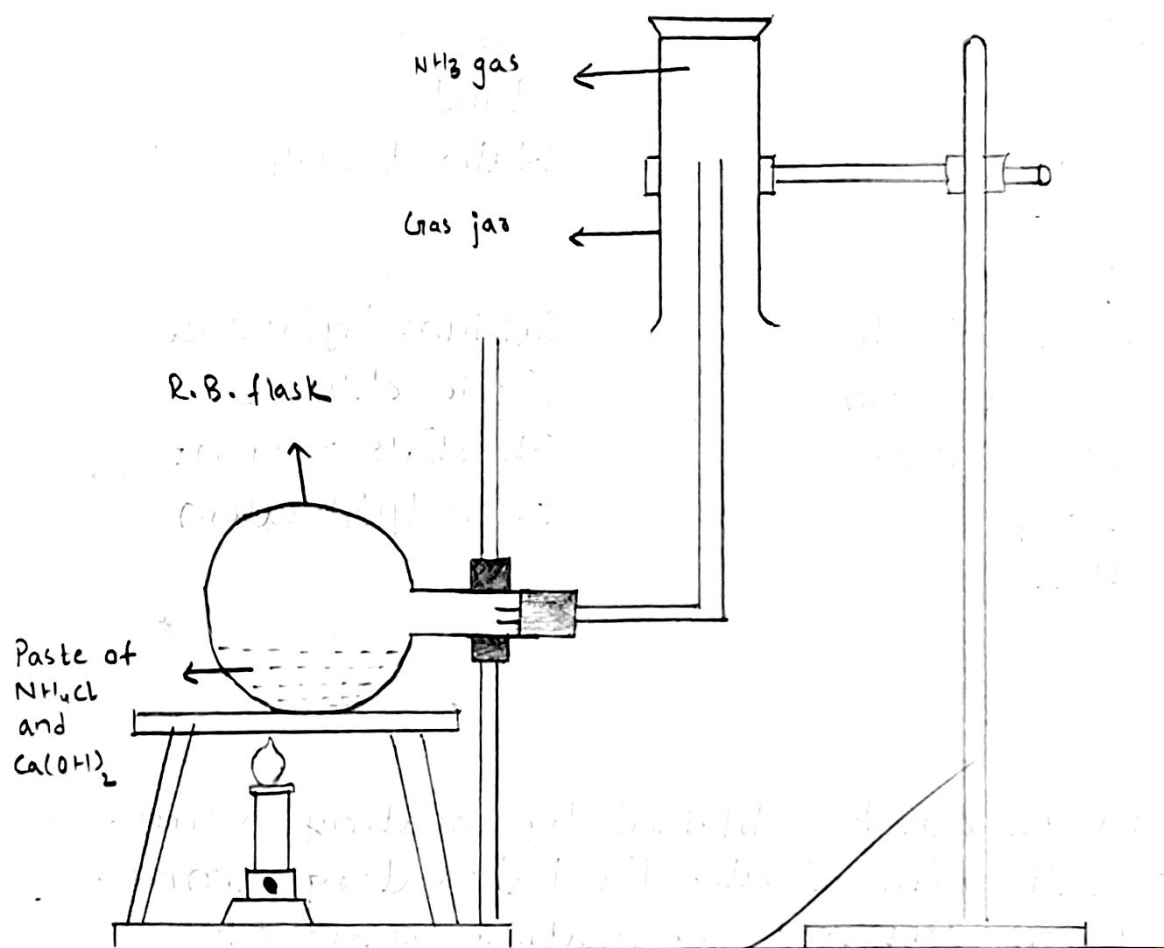


fig-laboratory preparation of ammonia gas

forward reaction. As ammonia gas is lighter than air and highly soluble in water it is collected by downward displacement of air. Paste of NH_4Cl should be used because NH_4Cl is volatile and sublimes on heating.

If dry ammonia is required then the moist ammonia gas is passed through a tower containing quick lime (CaO) which absorbs moisture present in ammonia.

PROCEDURE

A paste of Ammonium chloride and slaked lime was prepared in the ratio of 1:3 with little water in R.B flask. The apparatus was fitted airtight as shown in the figure and the flask was clamped in a slightly inclined position. Then the flask was heated gently, when ammonia gas formed it was collected by the downward displacement of air.

OBSERVATION TABLE

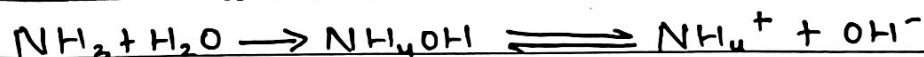
S.N.	Experiments	Observations	Inferences
①	The color and odor of gas was observed.	No color was seen but sharp pungent odour was observed.	NH_3 gas is a colorless gas having sharp pungent odour.
②	Solubility test The test tube filled with ammonia gas was inverted over the water in water trough.	Water level was rise in test tube rapidly	NH_3 gas is highly soluble in water.
③	Combustibility test A burning match was introduced into the gas jar filled with NH_3	Burning match stick extinguished .	NH_3 gas is non-combustible and non-supporter of combustion.
④	Litmus test Moist red and blue litmus papers were introduced into the gas jar containing ammonia gas.	Red litmus paper was turned into blue but blue litmus paper was remained unchanged.	Moist NH_3 gas is alkaline in nature due to the formation of NH_4OH
⑤	Phenolphthalein test - NH_3 gas was passed through water containing 1 to 2 drops of phenolphthalein in test tube.	Color of solution was turned pink.	NH_3 soln is alkaline in nature due to the formation of NH_4OH

S.N.	Experiments	Observations	Inferences
⑥	Reaction with conc. HCl The glass rod dipped in conc. HCl near the NH_3 gas was exposed.	Dense white fume was appeared.	Formation of NH_4Cl .
⑦	Reaction with mercurous nitrate The paper soaked in mercurous nitrate soln near NH_3 gas was exposed.	Paper was turned black.	Formation of mixture of black metallic mercury & mercuric amino nitrate.
⑧	Run with CuSO_4 soln - NH_3 gas through 2ml CuSO_4 soln in a test tube was passed till excess.	Bluish white PPT was appeared first which later turned into deep blue in excess of NH_3 .	Formation of bluish white $\text{Cu}(\text{OH})_2$ which finally forms deep blue complex of tetraammine cupric sulphate.
⑨	Run with Nessler's reagent NH_3 gas was passed through Nessler's reagent in test tube.	Brown ppt was appeared.	Formation of Million's base.
⑩	Run with FeSO_4 soln NH_3 gas was passed through FeSO_4 soln in test tube.	Dirty green PPT was appeared.	Formation of ferrous hydroxide.

⑪ FeCl_3 test NH_3 gas was passed through FeCl_3 soln in a test tube	Reddish brown ppt was appeared.	Formation of ferric hydronide.
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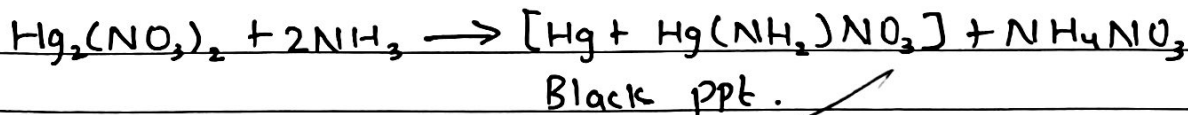
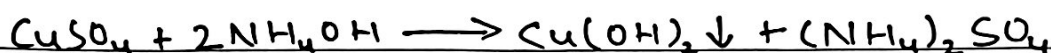
TEST REACTIONS

1. Reaction with water

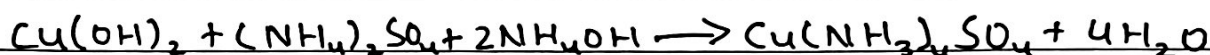


Moist Red Litmus + $\text{OH}^- \rightarrow$ Blue Litmus

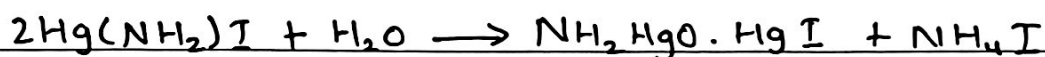
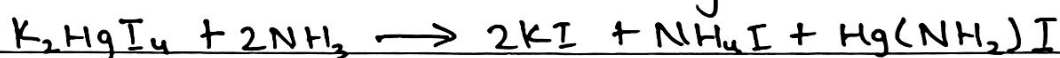
2. Reaction with Mercurous nitrate

3. Reaction with CuSO_4 solution

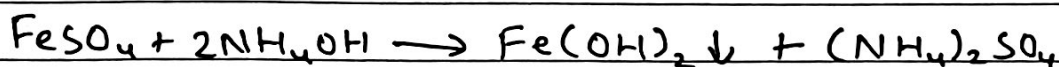
Bluish white Ppt



4. Reaction with Nessler's reagent



Reddish brown Ppt.

5. Reaction with FeSO_4 solution

Dirty green ppt.

6. Reaction with FeCl_3 solution

RESULT

In this way, ammonia gas was prepared in laboratory and its properties are studied.

CONCLUSION

Hence from above experiment it is concluded that ammonia gas can be prepared by heating the mixture of ammonium salt and alkali.

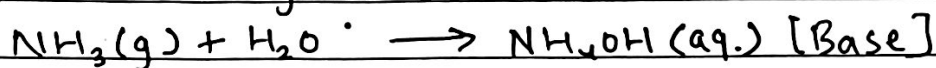
PRECAUTION

1. Personal protective equipment should be worn.
2. The R.B. should be inclined as shown in figure.
3. Excess water should be avoided while making paste.
4. The apparatus should be air tight.

VIVA-VOCE

1. Why is ammonia basic in nature?
2. What is oxidation number of N in NH_3 , N_2O , NO , and NO_2 & HNO_3 ?
3. Why dry mixture of NH_4Cl and $\text{Ca}(\text{OH})_2$ cannot be used for the preparation of ammonia?
4. Why is conc. H_2SO_4 is not used to dry the ammonia gas?

1. Because ammonia reacts with water to produce an ammonium hydroxide base.



2. -3, +1, +2, +4 and +5 respectively.

3. Because dry ammonium chloride gets converted into vapor form and may condense on cooler part of flask as sublimate and dry powder of $\text{Ca}(\text{OH})_2$ may convert into CaO while heating. This may cause the less or no production of ammonia gas.

4. Conc. H_2SO_4 is not used to dry ammonia gas because it reacts with ammonia to form ammonium sulphate $(\text{NH}_4)_2\text{SO}_4$. Since ammonia is absorbed and reacts instead of just losing moisture, it cannot be dried using conc. H_2SO_4 .